

Tmem27: a cleaved and shed plasma membrane protein that stimulates pancreatic beta cell proliferation.

SCIFACT-EVIDENCE-24142891

The signals and molecular mechanisms that regulate the replication of terminally differentiated beta cells are unknown. Here, we report the identification and characterization of transmembrane protein 27 (Tmem27, collectrin) in pancreatic beta cells. Expression of Tmem27 is reduced in Tcf1(-/-) mice and is increased in islets of mouse models with hypertrophy of the endocrine pancreas. Tmem27 forms dimers and its extracellular domain is glycosylated, cleaved and shed from the plasma membrane of beta cells. This cleavage process is beta cell specific and does not occur in other cell types. Overexpression of full-length Tmem27, but not the truncated or soluble protein, leads to increased thymidine incorporation, whereas silencing of Tmem27 using RNAi results in a reduction of cell replication. Furthermore, transgenic mice with increased expression of Tmem27 in pancreatic beta cells exhibit increased beta cell mass. Our results identify a pancreatic beta cell transmembrane protein that regulates cell growth of pancreatic islets.